

Practical Optimization Strategies for Coal-Washing Plants

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Abstract

The MODSIM mineral processing plant simulator is used to find optimal combinations of target separation specific gravities for the separation units in a typical coal washing plant. The plant is set up to wash three size fractions separately in parallel streams. Two alternative flowsheets are examined: the first includes a dense-medium vessel, a dense-medium cyclone and a 2-stage water-only cyclone, and the second includes a dense-medium vessel, a shaking table, a spiral separator and a bank of flotation cells. A simple graphical construction is used to find the optimal separation densities using data that is generated by the simulator. Results are presented as optimal ash content - total yield curves for the combined clean coal products.

Keywords: simulation, plant optimization, dense-medium separation, water-only cyclone.

Introduction

The calculation of the performance of coal washing units that are in common use is not difficult and the principles of making such calculations have been known for many years. Because the nature of individual coal particles do not change during passage through the various washing units in the plant, it is possible to calculate the performance of the whole plant by appropriate combinations of the calculation procedures for each unit operation. When the flowsheet includes comminution operations, the computations are somewhat more complex since the size distribution and washability distributions are transformed during the size-reduction processes. The population balance method describes both comminution and washing units correctly and forms the basis of simulation systems for coal washing and other ore dressing plants. The application of the population balance method to entire complex flowsheets can be formulated generally using a matrix that can be readily set up to include the topology of the flowsheet structure as well as a classification of units into three classes: separation units, transformation units, or water separation units [1]. In practice the generalized matrix formulation is seldom used for the simulation of ore dressing and coal washing

flowsheets since sequential iterative procedures have been found to be more robust and convenient particularly for allowing rapid addition or modification of models for the unit operations. This is particularly important because models are developed and modified continually as a result of ongoing research efforts.

Simulators have been available since the early 70's [1,2,3,4]), but these early programs were unwieldy and cumbersome to use, requiring data input through punched card medium and no facilities were provided for interactive specification of flowsheet structure and data and no graphical output of results was possible except graphics that were crudely produced by line printers. The situation changed radically in the late 70's with the introduction of good video-graphic terminals that allowed effective interaction between the user and the software, even on large mainframe computers. The early coal washing plant simulators had by then been fully documented and these could provide effective simulations [5, 6]. The introduction of the desk-top computer in the early 80's provided further significant enhancement of interactive and graphics capabilities and, perhaps more importantly, put significant computational power in the hands of engineers in a form that was convenient and easy to use. The demand for simulation systems encouraged research in their development [7] and there are now several excellent general-purpose modular simulation systems that are readily available which can be used to simulate the steady state performance of even the most complex mineral or coal processing plants.

Coal washing plants generally have comparatively simple flow sheet structures and the principles for the calculation of the performance of individual units in isolation and in simple combinations are now well established. Soon after the introduction of computers their potential for predicting washing plant performance was being exploited. It is now possible to simulate even the most complex coal washing flowsheets with accuracy that exceeds the demands of routine engineering analysis. Simulation is a particularly suitable technique to establish practical and useful set-up points for coal washing plants that include several different sub circuits to clean coals of various sizes. This is a common problem because different unit operations are used to handle the different size ranges and the optimal separation density varies with both particle size and equipment type. A plant that cleans the full size spectrum of sizes in a number (usually two or three) of subsections must be setup to optimize the combined operation. This paper shows how this optimization can be done using the MODSIM plant simulator together with a simple graphical construction..

Optimization of coal washing plants using models for the individual units has received attention in the literature and a variety of computational methods has been tried [8, 9, 10]. The graphical

procedure developed here has advantages over previously described methods and works equally well with both simulated and experimental data.

Plant Structure and Optimal Setup.

The flowsheet for the plant is shown in Figure 1 which is fairly typical of modern coal washing plants that treat a wide size-spectrum feed. Brief details of the models used to describe the equipment items in the plant are given in Table 1. The coarse material is cleaned in a dense-medium drum, the intermediate size in a dense-medium cyclone and the fines in water-only cyclones. In order to set up a plant such as this optimally, it is necessary to make five key decisions: the sizes at which the feed should be cut and the target specific gravities at which the dense-medium drum, the dense-medium cyclone and the water-only cyclone should be operated to get the best combination of ash rejection and coal yield. It turns out to be convenient to consider the choice of specific gravities and the choice of separation sizes as two separate optimization problems. Optimal settings for the target specific gravities can be obtained and then the optimal cut sizes can be computed. This paper deals with the first of these problems and we demonstrate that optimal combinations of the specific gravities can be easily obtained by simulation. The choice of the optimal settings for the unit separation densities can improve yields significantly by comparison with plants that are set up arbitrarily. Yield losses as high as 3 to 4% with non-optimal settings have been reported in the literature [8, 9]. Losses of that magnitude impact the economics of coal production significantly and should never be allowed to occur simply because the optimal combination of separation densities is not properly established.

Models for Coal Washing Unit Operations

The flowsheets to be simulated consist primarily of coal washing units each with a target specific gravity of separation. The feed is allocated to each washing unit on the basis of separation by size in the screening operations. This is a fairly common arrangement for coal washing plant flowsheets. The target specific gravity of separation for a unit is the separation point to be expected when the partition curve of the composite feed to clean coal is plotted without considering each narrow size fraction separately. All gravity separation units are known to show significant variation of separation performance with particle size. In general the cut point increases with decreasing particle size and the efficiency of separation decreases with decreasing particle size. This is particularly noticeable at particle sizes smaller than a few millimeters, the so-called fine coal separations. The spiral

separator deviates from this rule and shows a distinct minimum in the graph of cut point against particle size [10,11,12].

In order to calculate the quality of the clean coal and the discard from any unit a simulator must correctly model the variation of both cut point and separation efficiency with particle size. The particle size distribution and the distribution of particle densities in the feed to the unit must be known.

In general the cut point can be controlled by variation of an operational control in the plant or by means of a physical adjustment to the equipment. Dense-medium separators are set by controlling the medium density and the separation density in autogenous separators such as water-only cyclones, concentrating tables and jigs is controlled by physical adjustments. At the highest level, simulation models should accept as input data a specification of the equipment type and size and the settings of all controllable physical variables and be capable of predicting the cut points and separation efficiencies that will be achieved in practice. Models at this level are not always available and usually require very careful calibration against pilot or full-scale plant data.

In the present case no suitable data are available to calibrate high-level models and more restricted models which accept a target cut-point are used. The concept of the target cut-point originates with Gottfried and Jacobsen [13] who generalized the operating partition data that was presented over a number of years in the well-known series of USBM RI's that covered all the important coal cleaning equipment types. The method, which we refer to as the Gottfried-Jacobsen (G-J) procedure, establishes a correlation between particle size and the ratio of cut point for a single narrow size fraction to the overall cut-point.

In MODSIM, this procedure is generalized somewhat by defining the relationship as

$$r = \frac{SG_s}{\overline{SG_s}} = f\left(\frac{d_p}{\overline{d_p}}\right) \quad (1)$$

where SG_s is the cutpoint for particles of size d_p , $\overline{SG_s}$ is the overall cut point for the composite feed and $\overline{d_p}$ is the average particle size in the feed. This normalization of the G-J procedure with respect to average particle size in the feed is quite robust and correlates much of the data available in the literature and the function $f(\bullet)$ is not very sensitive to the feed composition and allows, at least approximately, for fairly wide variations in the size distribution in the feed. $\overline{SG_s}$ in equation (1)

is the target cutpoint and when G-J models are selected in MODSIM, the user must specify this target to fix the operating point of the particular unit. It has been pointed out [14], [15] that this procedure should be undertaken in an iterative manner since the actual separation density achieved for the composite feed will depend on the composition of the feed that enters the unit. The difference between the target cut point and the cut point that is actually achieved is usually quite small and iterative correction is not usually justified.

Plant Performance and Optimization

Before the plant can be optimized, it is necessary to investigate the performance of each of the individual washing units under conditions that each one experiences in the flowsheet. This is the first task for the simulator which can be used to investigate the range of performance of each washing unit in the plant. The performance of each washing unit in the flowsheet in Figure 1 was simulated over a range of target cut-points. The results are summarized in Figure 2. The figure shows the cumulative yield against the cumulative ash for each of the three washing units together with the theoretical yields calculated from the washability data of the feed material to each unit.

The arrangement of three separation units in parallel lends itself to optimization so that the plant can produce clean coal at the greatest possible yield at a given ash content or at the minimum ash content for a given yield. The optimal combination of target cut-points for the three washing units is easy to establish in this case using the simulated data from Figure 2. The calculated yields are replotted as the product of yield $\times$ ash content vs. yield for each of the three units as shown in Figure 3. Each calculated point is indexed by the target cut-point in the unit. Optimal combinations of the three target cut-points for the individual washing units are located at points on these curves where the slopes of the curves are identical. Seven such optimal combinations are shown as points connected by broken lines in Figure 3. A typical set of parallel lines that touch the three curves at points where the slopes are equal are also shown. The appropriate target cut-points can be established by interpolation between the target cut-points that label each calculated point on the curves. The 7 optimal target specific gravity combinations are given in Table 2 together with the optimal yields, ash content and energy content of the clean coal product. The plant can be set up optimally for any preselected ash or energy content by interpolation between the values given in Table 2. It is interesting that the optimal sets of target cut-points do not depend on the relative flow rates to the three washing operations. The theoretical justification for the graphical optimization procedure is set out in Appendix 2.

The performance of the plant at the seven optimal combinations of target cut points is shown in Figure 4 where the optimal plant performance is compared to the extreme performance as defined by the washability of the feed. This extreme performance is hypothetical and would be achieved only if each washing unit could be operated at perfect separation efficiency and if the three units were set up at the optimal combination of cut points. The optimal performance is reflected also in the energy content of the cleaned coal that can be produced when the plant is set up optimally. The optimal recovery of energy content is shown as a function of yield in Figure 5. The results obtained show just how useful simulation techniques are for the solution of plant optimization problems even at the elementary level of modeling that is used here.

The potential loss of yield that can result when a plant is set up without regard to the optimal combination of cutpoints can be gauged from the following observation. The problem that is considered in this paper was submitted to the author as a standard problem for demonstration and comparison of coal washing plant simulators at COAL PREP 94 [16]. It was recommended that simulations should be based on the following combination of set points: dense-medium drum 1.35, dense-medium cyclone 1.65 and the water-only cyclone 1.95. The performance of the simulated plant with this combination of cutpoints is shown in Figures 4 and 5. It is immediately clear that this combination would lead to grossly inefficient operation and should not seriously be considered for any plant that has units operating as described by the models used here.

Optimizing the Plant with Alternative Washing Units

In addition to the selection of optimal cut points for the washing units, simulation is an ideal tool for the investigation of alternative washing unit operations without the need for expensive and time-consuming plant or field experimentation. A number of alternative techniques can be considered for each of the three process streams in the plant. An alternative choice of units for the plant is shown in the flowsheet in Figure 6. A shaking table is proposed to treat the intermediate size and the fines are classified further using a Derrick Screen to produce an ultra fine stream for final cleaning by froth flotation and a fine stream for cleaning in a spiral. Obviously it makes sense to compare overall plant performance at optimal settings when a choice is to be made regarding flowsheet alternatives. Because the simulator is so easy to use, any combination of alternative units can be quickly and cost-effectively investigated with the consequent reduction in cost and significant improvement in plant productivity which results when the plant is set up correctly with the units chosen.

The concentrating table was modeled using a Gottfried-Jacobsen model and the spiral was modeled using an updated version of the model developed by King *et. al.* [11]. This model is based on data obtained in a laboratory study of a typical coal washing spiral. The model allows for the joint effect of both size and density on behavior of particles in the spiral. The flotation stage was modeled using discrete distribution of flotation rate constants with allowance for bubble loading limitations which is particularly important for coal flotation [17]. Typical values for the specific flotation rate constants and their distribution for each washability fraction were chosen for purposes of this exercise. These parameters are normally estimated from well-defined laboratory batch or continuous pilot- or full-scale plant tests.

The performance of each of the three washing units with respect to their own feeds is shown in Figure 7. The concentrating table was controlled by variation of the target cutpoint. The performance of the spiral was controlled by variation of the relative cutter position at the discharge end, the effective cut point increasing the closer the cutter is set to the center column of the spiral. Control can be exercised over the flotation operation by variation of the aeration rate although the scope of this variation is not great.

The optimal combination of table, spiral, and flotation control setting were obtained using the optimization procedure that was used for the original flowsheet. The construction is shown in Figure 8. The optimal setting for the dense-medium drum was obtained by transferring the slope of the tangent lines to Figure 3 and finding the point on the dense-medium drum line that was tangential to this tangent line. The optimal plant performance is shown in Figure 4 and a summary of the simulated performance is given in Table 3.

The data presented in Figure 4 shows again that significant gains in plant productivity can be made by setting up the alternative units optimally with respect to each other. However, the optimal yield - ash curve for the alternative flowsheet falls slightly below that for the original flowsheet. Consequently the more complex alternative flowsheet is unlikely to be attractive for this coal washing operation and need not be considered further.

Discussion

The simple solution to an important practical operating problem given here immediately raises the question of reliability of the solution. Can the results obtained here be applied to a particular operating coal washing plant and optimal performance realized? The answer must be negative since that would expect too much of the models used in the simulator. The models would first have to

be calibrated against operating data that is known to be relevant to the specific units as they operate in the plant. This would not be a particularly onerous task and is routine for the application of simulation techniques in mineral processing plants.

Of course, the graphical method that is proposed here for the selection of optimal operating cutpoints for the individual units can be based entirely on experimental data because the simulated points in Figures 2 and Figure 7 can be determined from sampling the feed and products of each of the units as they operate in the plant. However, this would be costly and time consuming to implement and would necessitate the operation of the plant at non-optimal settings for significant time periods while the data were being generated. The data must generate sufficiently smooth curves when plotted as in Figures 3 and 6 to allow for the reasonably accurate positioning of the tangent lines. The use of simulation with calibrated models obviously offers a far more economical and practical method. It is also possible to take a more analytical approach and base the optimization calculation on the differentiation of the partition curve [18] once the feed to each of the units has been characterized by the simulator. Thus simulation offers a number of variations that can be adopted by the engineer to select optimal operating points to meet appropriate technical and economic objectives for the plant in question.

The method developed here can be used on a continuing basis with optimal settings determined as frequently as required to allow the plant to perform optimally in the face of varying characteristics of the plant feed. In particular MODSIM can simulate the effect of varying liberation characteristics of ash and pyrite and these can have a profound effect on plant performance. This will be the subject of a future publication.

Conclusions

Simulation using a versatile general - purpose simulator equipped with realistic models for the unit operations has been shown to be an effective technique for the determination of optimal cut points for dense - medium and other gravity separation and flotation units in a coal washing plant.

The graphical method used to establish optimal combinations of cut points for the washing units is quick and simple to use once the necessary data has been generated by the simulator.

Simulation is an effective tool for the selection of alternative unit operations and flow sheet configuration without the need to undertake lengthy and costly plant trials.

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APPENDIX 1: RAW COAL PROPERTIES USED IN THE SIMULATOR

RAW-COAL SIZE ANALYSIS

Size		Weight (Wt%)	Ash (Wt%)
Passed	Retained		
	1 1/2 IN	15.8	18.12
1 1/2 IN	3/4 IN	22.5	13.44
3/4 IN	3/8 IN	18.2	12.17
3/8 IN	28M	33.2	11.08
28M	100M	4.5	10.88
100M	200M	1.7	12.25
200M		4.1	27.07

RAW-COAL
WASHABILITY ANALYSIS

Specific Gravity			Ash (Wt%)	Sulfur (Wt%)	Heating Value (Btu/lb)
Sink	Float	Representative value			
	1.30	1.303*	2.96	0.99	14812
1.30	1.35	1.325	5.93	1.13	14287
1.35	1.40	1.375	11.04	1.83	13350
1.40	1.45	1.425	16.07	1.68	12502
1.45	1.50	1.475	22.78	1.26	11424
1.50	1.60	1.550	28.99	0.91	10386
1.60	1.70	1.650	36.43	0.83	9122
1.70	1.80	1.750	45.71	0.58	7493
1.80	2.17	1.985	58.99	0.88	5296
2.17		2.534*	81.52	2.38	1850

* Estimated from mineral vs (specific gravity)⁻¹ plot as described below.

Specific Gravity		1 1/2 X	3/4 X	3/8 X	28M X	100M X	200M*
Sink	Float	1 1/2 Weight (Wt%)	3/4 Weight (Wt%)	3/8 Weight (Wt%)	28M Weight (Wt%)	100M Weight (Wt%)	200M Weight (Wt%)
	1.30	37.7	42.4	41.0	50.3	63.0	60.7
1.30	1.35	31.1	28.0	30.1	23.4	14.2	12.9
1.35	1.40	1.9	7.4	9.0	8.1	5.9	4.0
1.40	1.45	1.4	2.9	4.4	4.3	3.2	4.9
1.45	1.50	2.4	2.0	1.9	1.7	1.4	2.1
1.50	1.60	4.0	5.4	3.6	2.4	2.2	2.9
1.60	1.70	3.7	3.4	2.4	1.7	1.3	1.7
1.70	1.80	2.9	1.9	1.8	1.4	0.9	1.0
1.80	2.17	9.0	4.0	3.3	2.8	2.4	2.7
2.17		5.9	2.6	2.5	3.9	5.5	7.1

* Estimated as described below.

DATA COMPLETION

The data as supplied was incomplete in two important respects and required completion. The effective densities of the two outer washability fractions were not given and the washability analysis of the 200 mesh × 0 size fraction was missing. These omissions are commonly found in coal washability data sets. The effective densities of the outer washability fractions determine the behavior of this material that is predicted by the unit models and a rational assignment of effective density is necessary particularly when the units will be set up to cut at either high or low specific gravities. The washability distribution of the 200 mesh × 0 fraction was estimated by assuming that

the lower 9 density fractions were distributed identically to the those in the 100 mesh $\times$ 200 mesh fraction and the percentage in the highest density fraction was adjusted to make the ash content of the 200 mesh $\times$ 0 fraction equal to the experimentally determined value which was supplied. The resultant distribution in the fine size fraction is shown in the table above.

The choice of the effective densities in the outer washability fractions was based on the fact that accurately determined values of the ash and sulfur content were available for these intervals. The ash and sulfur values were converted to mineral matter using the Parr formula

$$\text{Mineral matter} = 1.08 \times \text{Ash} + 0.55 \times \text{Sulfur}$$

The relationship between mineral matter and particle density is simple for many coal samples since coal can often be described as a two-phase mixture of two mineral types of different apparent relative densities usually about 1.28 for the combustible component and less than 3.0 for the mineral matter. Then mineral matter content will be a linear function of the reciprocal of the density or specific gravity. This relationship has been comprehensively explored for South African coals [21]. The data is plotted as mineral matter against the reciprocal average specific gravity of the interval in Figure 9. The linear dependence is clearly evident and the effective densities can be evaluated from the equation of the regression line through the points at the known mineral matter values of the outer fractions. In this way these fractions were assigned specific gravities of 1.30 and 2.53 respectively as shown in Figure 9.

APPENDIX 2: SELECTION OF OPTIMAL COMBINATIONS OF CUT POINTS

The yield of clean coal in a plant which has three separate washing units in parallel such as that shown in Figure 1 is equal to the sum of the yields from each section

$$Y = Y_1 + Y_2 + Y_3 \quad (\text{A1})$$

Where Y and Y_i represent total yield and yield in section i relative to the total tonnage in the feed. Equation A1 represents a total mass balance over the clean coal sump in Figure 1.

The ash content of the clean coal is calculated from

$$a = \frac{Y_1 a_1 + Y_2 a_2 + Y_3 a_3}{Y} \quad (\text{A2})$$

where a_i is the ash content of the clean coal from section i .

The operation of the plant is optimized if Y is maximized at a fixed clean coal ash value, a , or clean coal ash is minimized at a fixed value of the total yield, Y . Both of these optimization problems are solved at any operating condition at which $dY = 0$ and $da = 0$ simultaneously.

From equation A2

$$da = \frac{d(Y_1 a_1 + Y_2 a_2 + Y_3 a_3)}{Y} - \frac{a}{Y} dY \quad (\text{A3})$$

which requires that

$$d(Y_1 a_1 + Y_2 a_2 + Y_3 a_3) = 0 \quad (\text{A4})$$

at an optimal set of operating conditions. The product $Y_i a_i$ is a unique function of Y_i for each unit as illustrated in Figures 3 and 8 which means that

$$d(Y_1 a_1 + Y_2 a_2 + Y_3 a_3) = \frac{dY_1 a_1}{dY_1} dY_1 + \frac{dY_2 a_2}{dY_2} dY_2 + \frac{dY_3 a_3}{dY_3} dY_3 \quad (\text{A5})$$

The expression on the right side of equation A5 must be zero at an operating point at which dY is also zero. One way to guarantee this is to make each of the slopes $\frac{dY_i a_i}{dY_i}$ equal

$$\frac{dY_1 a_1}{dY_1} = \frac{dY_2 a_2}{dY_2} = \frac{dY_3 a_3}{dY_3} = m \quad (\text{A6})$$

then

$$\begin{aligned} d(Y_1 a_1 + Y_2 a_2 + Y_3 a_3) &= m(dY_1 + dY_2 + dY_3) \\ &= m dY \\ &= 0 \end{aligned} \quad (\text{A7})$$

Thus choosing the operating points to make the slopes equal as illustrated in Figures 3 and 8 is sufficient to ensure optimality.

Table 1 MODSIM models that were used in the simulation.

Unit	MODSIM model	Brief description	Model parameters
Single deck vibrating screen	SCRN	General purpose vibrating screen model based on capacity factors.	Mesh size 9.5mm
Sieve bend	SCR1	Model based on the partition factor with carry over of fines in proportion to the water that reports to the overflow	Cut point $d_{50} = 600\mu\text{m}$ Water split to undersize = 30%
Dewatering screen	DWSC	Model due to Ng [19], [20] is based on the rate of transport and kinetics of drainage.	Ultimate moisture content Amplitude of vibration Frequency of vibration Angle of inclination Mesh size Length of screen Width of screen Angle of vibration
Dense-medium drum	DRUM	Model based on partition function. Gottfried-Jacobsen procedure for cut point.	Target specific gravity of separation.
Dense-medium cyclone	DMCY	Model based on partition function. Gottfried-Jacobsen procedure for cut point.	Target specific gravity of separation.
Water-only cyclone	WOCY	Model based on partition function. Gottfried-Jacobsen procedure for cut point.	Target specific gravity of separation. Single- or two-stage

Table 2. Optimal settings for the washing units in the plant and the resulting plant performance.

Optimal combination of target specific gravities			Calculated plant performance				
Water-only cyclone	Dense-medium cyclone	Dense-medium vessel	Product	Ash content %	Yield %	Calorific value Btu/lb	Sulfur content %
1.31	1.37	1.39	Clean coal	4.89	69.9	33.66	1.10
			Discard	38.0		22.66	1.31
1.55	1.39	1.41	Clean coal	5.19	75.6	33.54	1.11
			Discard	38.6		20.47	1.32
1.65	1.41	1.44	Clean coal	5.51	79.2	33.41	1.12
			Discard	43.2		18.69	1.31
1.69	1.44	1.46	Clean coal	5.75	81.2	33.31	1.13
			Discard	46.1		17.58	1.29
1.73	1.47	1.50	Clean coal	6.09	83.3	33.18	1.14
			Discard	49.4		16.31	1.29
1.77	1.66	1.59	Clean coal	7.08	87.3	32.79	1.13
			Discard	56.5		13.57	1.36
1.86	1.91	1.84	Clean coal	8.63	91.72	32.18	1.12
			Discard	65.5		10.17	1.63

Table 3. Optimal settings for the washing units in the alternative flowsheet and the resulting plant performance.

Optimal combination of control settings				Calculated plant performance				
Flotation cells. Aeration rate m ³ /s per m ³	Spiral. Position of splitter	Concentrating table. Target sp. gravity	Dense- medium vessel target sp. gravity	Product	Ash %	Yield %	C.V. Btu/lb	S %
0.56	0.74	1.55	1.43	Clean coal	5.58	78.3	14350	1.12
				Discard	42.4		8173	1.29
0.60	0.72	1.57	1.45	Clean coal	5.73	79.9	14324	1.13
				Discard	44.8		7775	1.29
0.65	0.66	1.62	1.47	Clean coal	5.95	81.8	14284	1.13
				Discard	47.8		7280	1.29
0.72	0.64	1.66	1.505	Clean coal	6.21	83.4	14240	1.13
				Discard	50.5		6833	1.30
0.80	0.61	1.71	1.53	Clean coal	6.50	84.8	14195	1.13
				Discard	53.0		6414	1.32
0.84	0.59	1.75	1.56	Clean coal	6.77	85.9	14148	1.13
				Discard	55.1		6061	1.34
0.95	0.40	1.68	1.68	Clean coal	7.58	88.8	14007	1.12
				Discard	61.1		5087	1.45

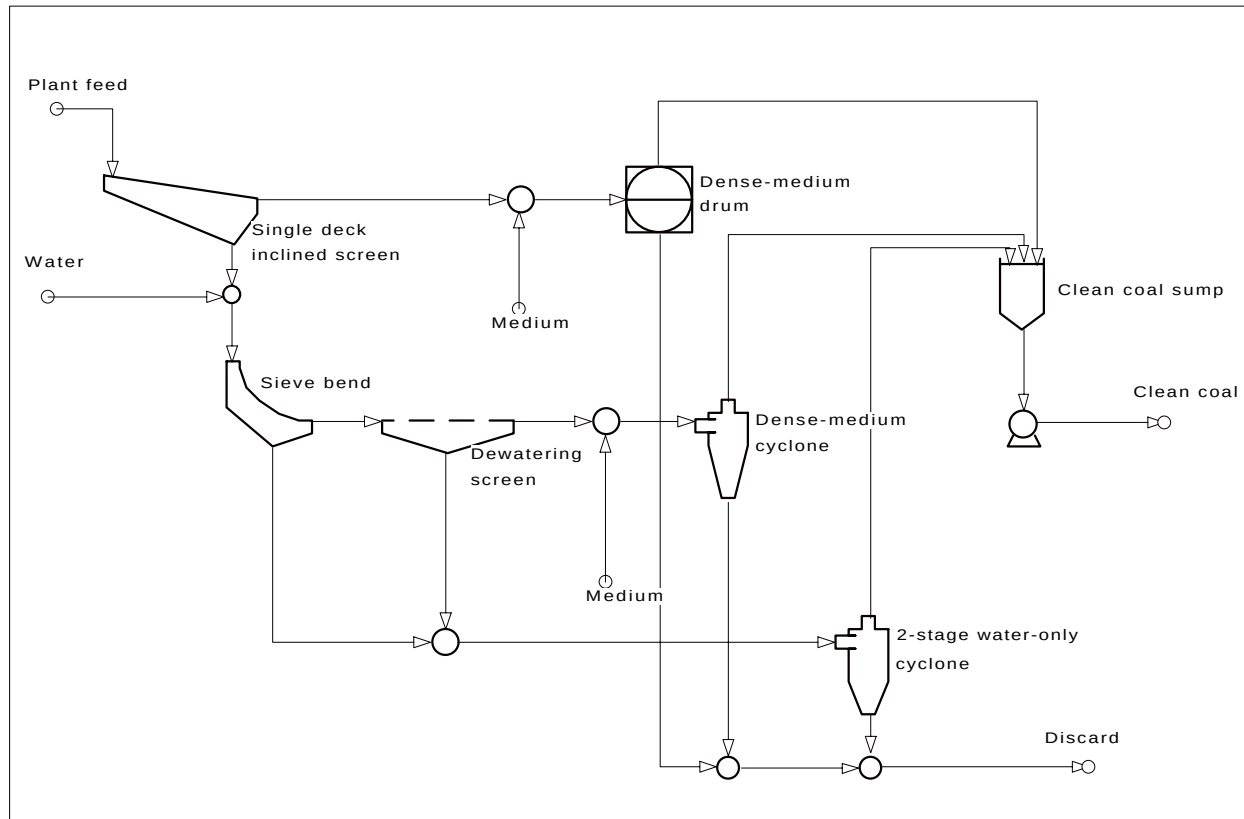


Figure 1 The flowsheet of the plant to be optimized by finding the optimum settings for the separation densities in the dense-medium drum, in the dense-medium cyclone and in the water-only cyclones. For clarity the medium recovery circuit is not included in the flowsheet.

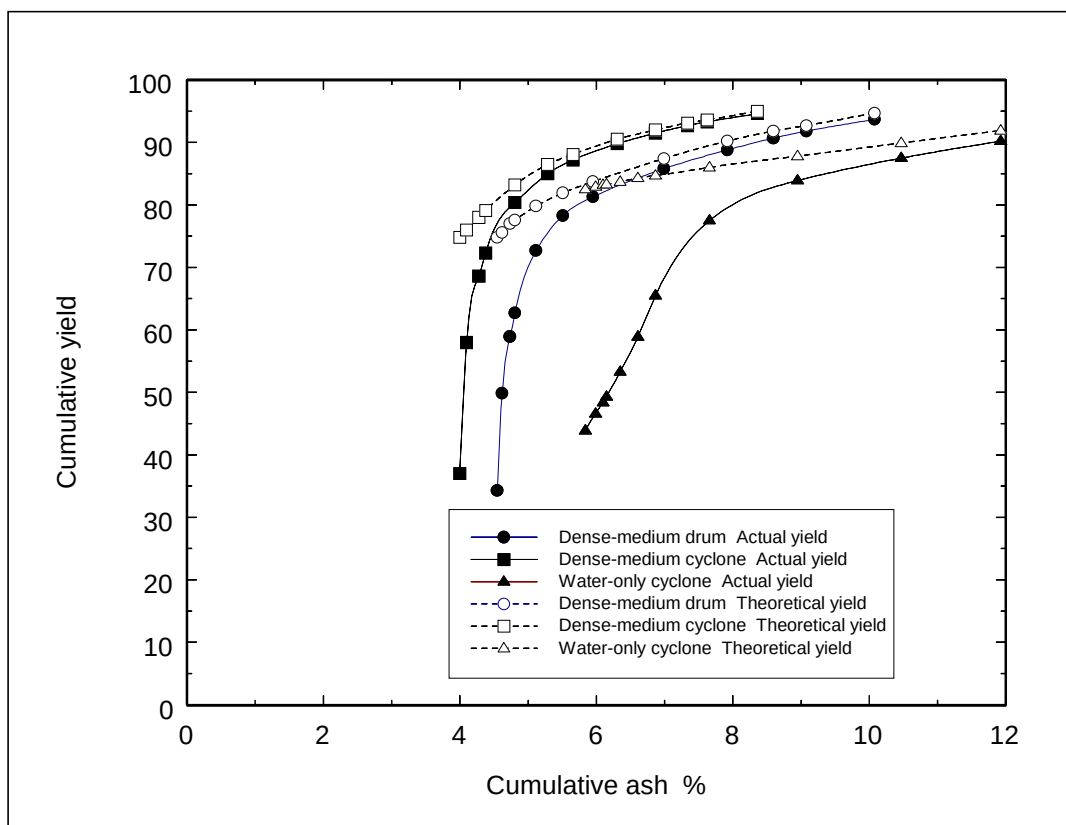


Figure 2 Performance curves for each of the three washing units compared to the washabilities of their respective feeds. Each point shown represents a different target specific gravity for each unit.

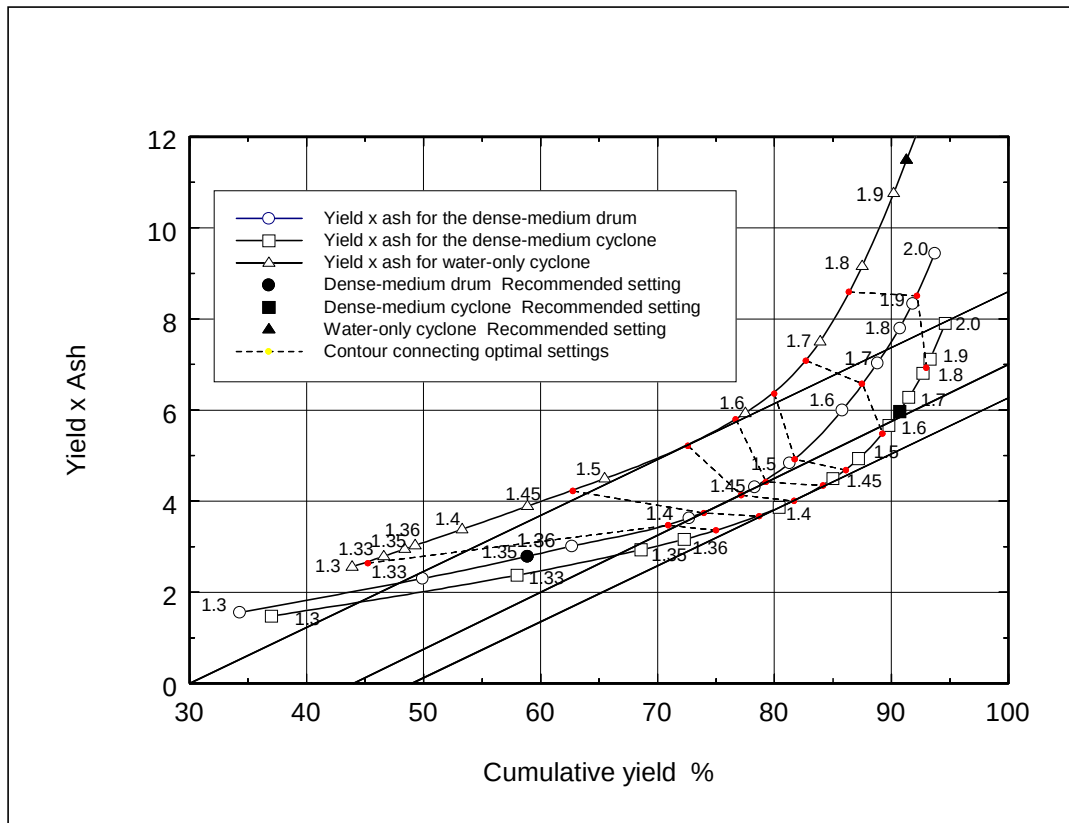


Figure 3 Ash-yield products for the three individual washing units. Points where these curves have equal slopes define optimal combinations of target specific gravities. Slopes at one particular optimal combination are shown.

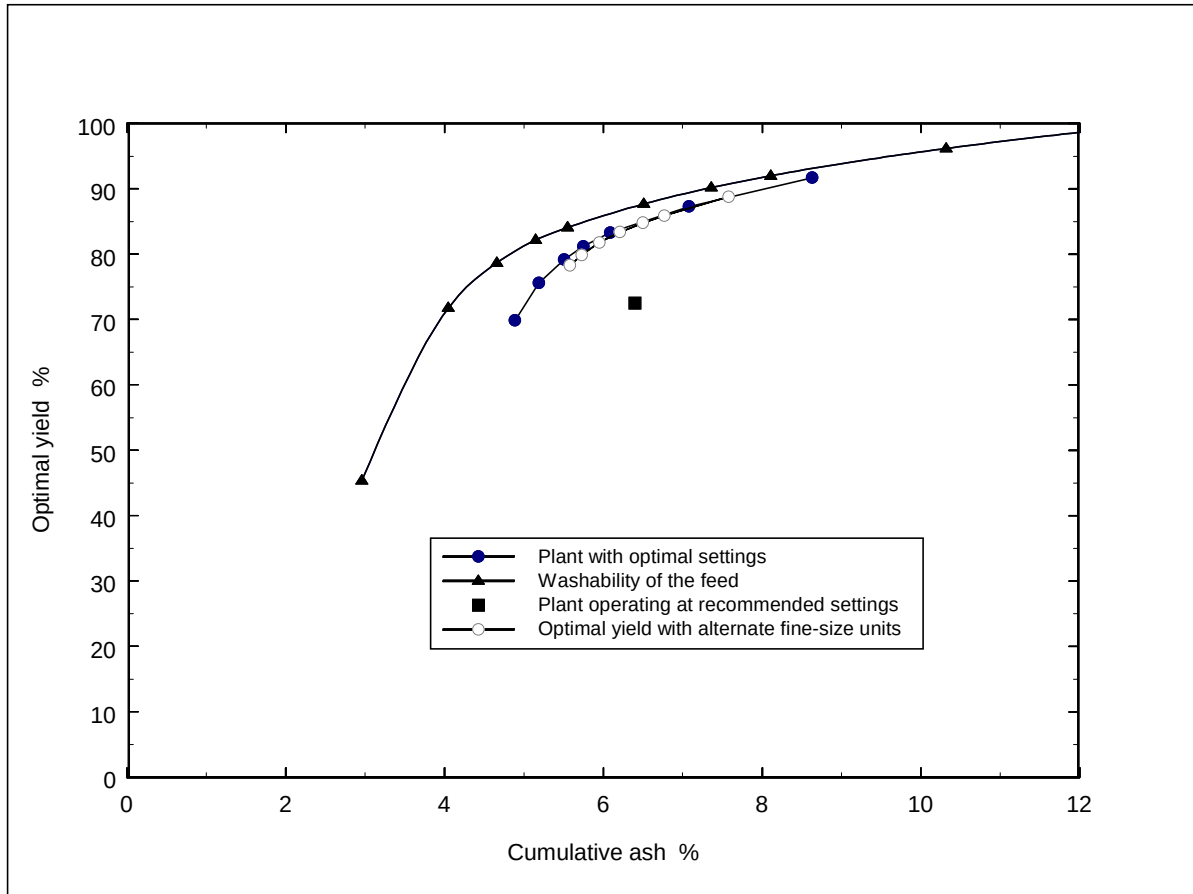


Figure 4 *Optimal performance curve for the plant compared to the washability of the feed and to the plant performance at the recommended settings. The washability of the feed represents the extreme limit of good plant performance and will be achieved only if each of the three separation units in the flowsheet is operated with perfectly clean separations and the optimal combination of cut points is used.*

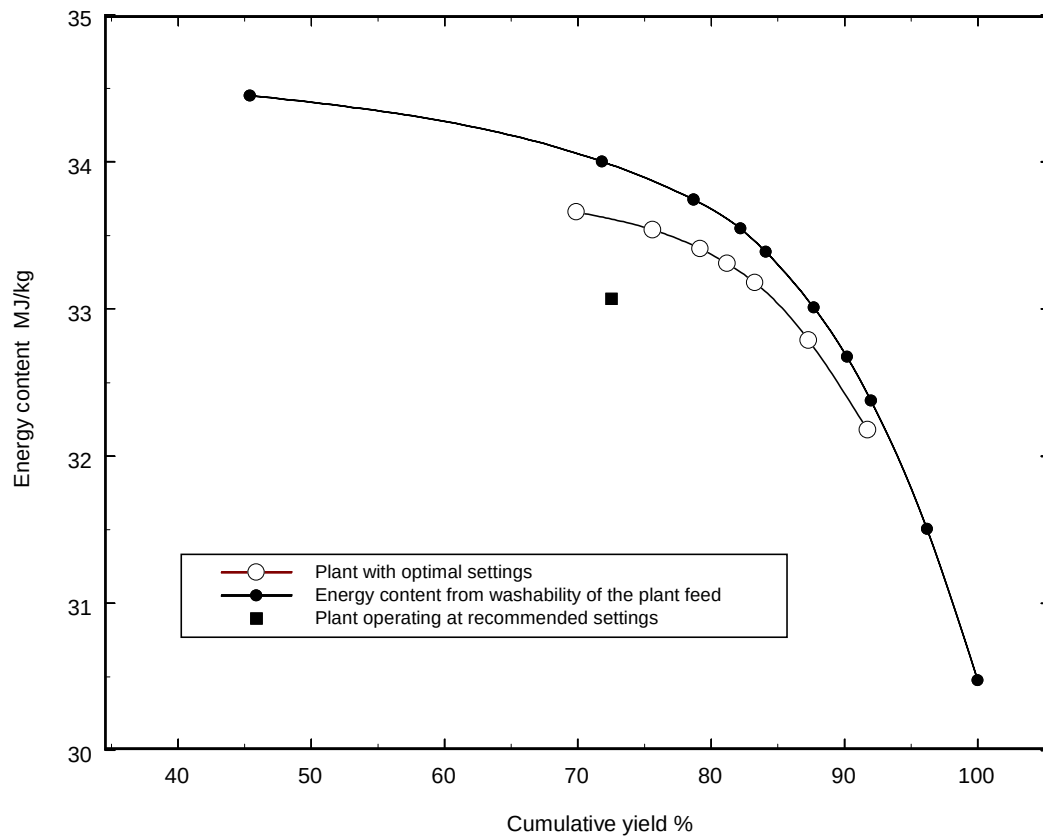


Figure 5 *Optimal performance curve for the plant compared to the ideal and to the plant operated at the recommended settings.*

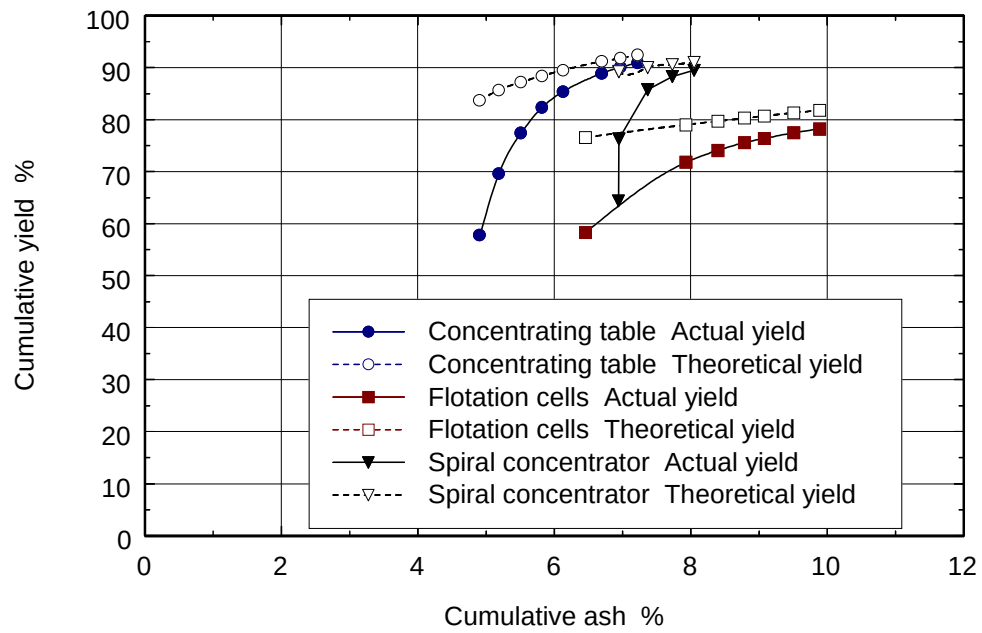


Figure 6 *Performance of each of the alternative fine-coal separation units relative to their respective feeds*

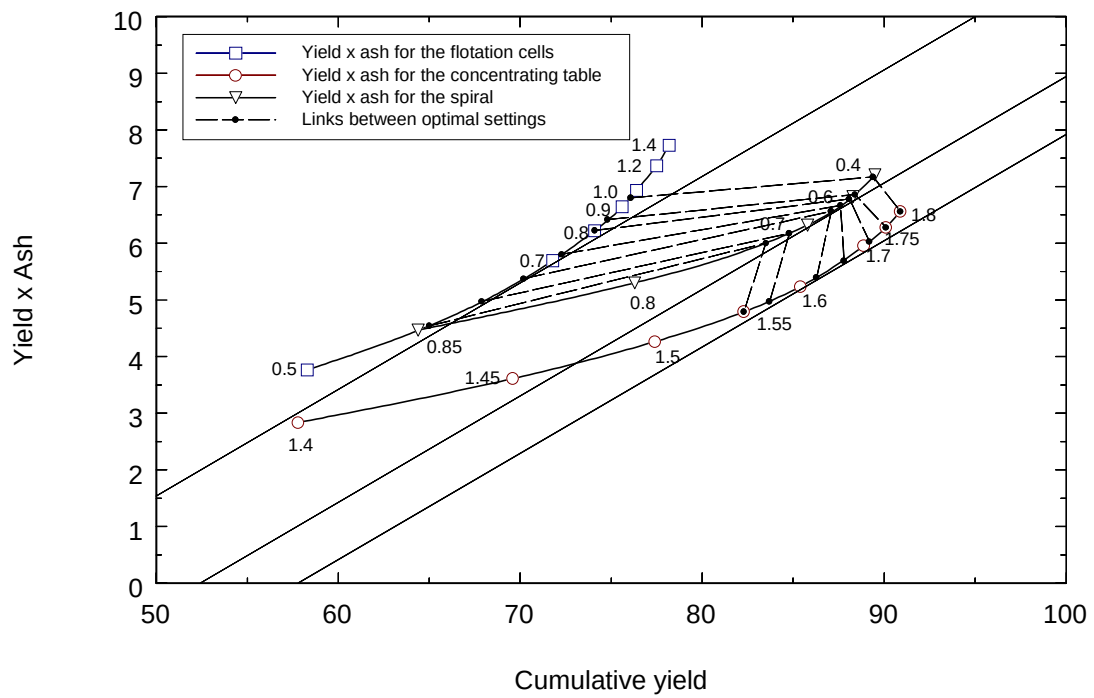


Figure 7 Construction showing how to select optimal combinations of the control settings for the washing units in the alternative flowsheet shown in Figure 6.

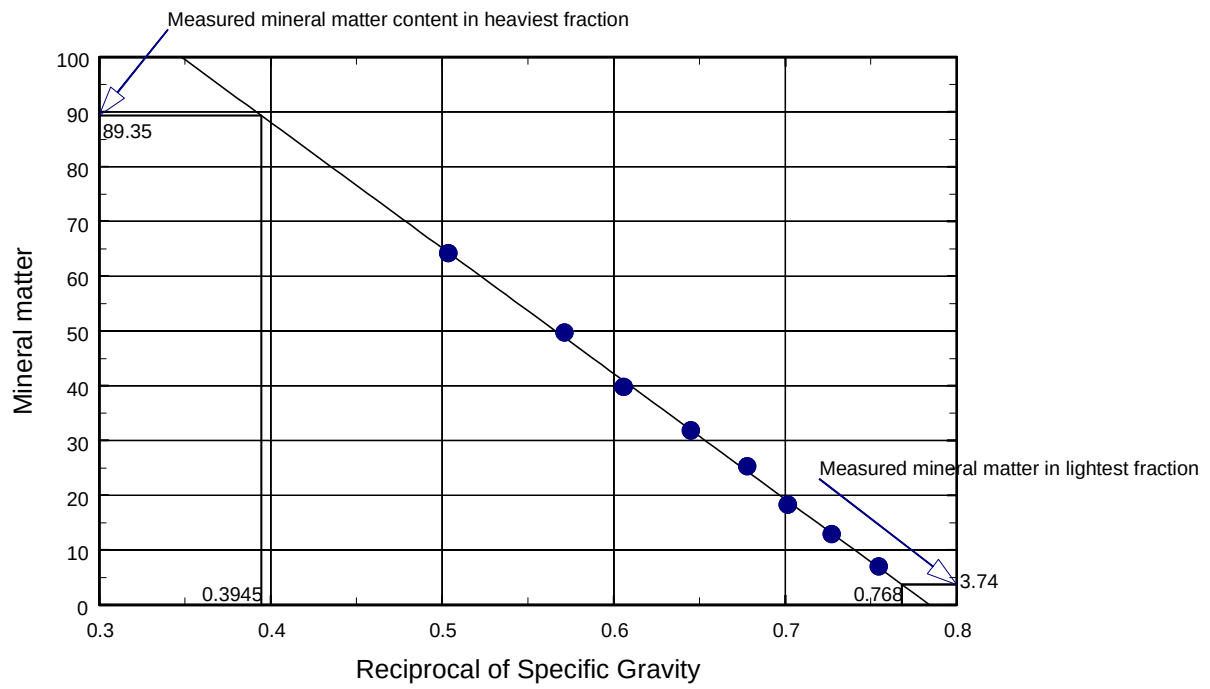


Figure 8 Extrapolation to find the effective density of the two outer washability fractions.